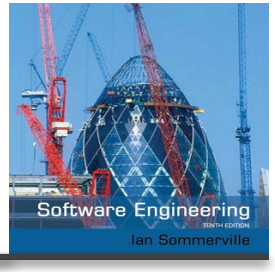
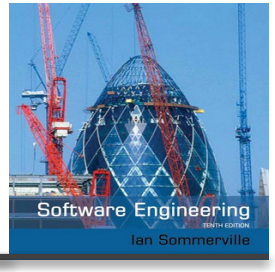
Essential attributes of good software



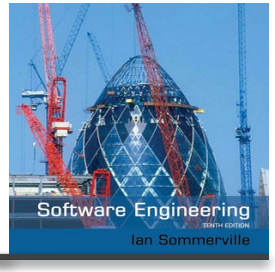
Product characteristic	Description
Maintainability	Software should be **written in such a way so that it can evolve to meet the changing needs of customers**. This is a critical attribute because software change is an inevitable requirement of a changing business environment.
Dependability and security	Software **dependability includes a range of characteristics including reliability, security and safety**. Dependable software should not cause physical or economic damage in the event of system failure. Malicious users should not be able to access or damage the system.
Efficiency	Software should not make wasteful **use of system resources such as memory and processor cycles.** Efficiency therefore includes responsiveness, processing time, memory utilisation, etc.
Acceptability	Software must be acceptable to the type of users for which it is designed. This means that it must be **understandable, usable and compatible with other systems that they use.**

Software engineering



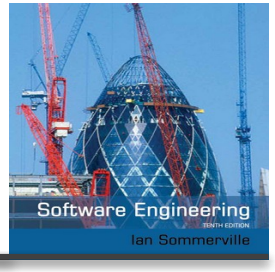
- Software engineering is an engineering discipline that is **concerned with all aspects of software production from the early stages of system specification through to maintaining the system after it has gone into use.**
- Engineering discipline
  - Using appropriate theories and methods to solve problems bearing in mind organizational and financial constraints.
- All aspects of software production
  - Not just technical process of development. Also project management and the development of tools, methods etc. to support software production.

Importance of software engineering



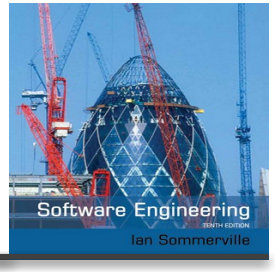
- More and more, individuals and society rely on advanced software systems. We need to be able to produce reliable and trustworthy systems economically and quickly.
- **It is usually cheaper, in the long run, to use software engineering methods and techniques for software systems rather than just write the programs as if it was a personal programming project.**
  - **For most types of system, the majority of costs are the costs of changing the software after it has gone into use.**

Software process activities



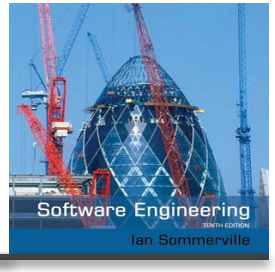
- **Software specification**, where customers and engineers define the software that is to be produced and the constraints on its operation.
- **Software development**, where the software is designed and programmed.
- **Software validation**, where the software is checked to ensure that it is what the customer requires.
- **Software evolution**, where the software is modified to reflect changing customer and market requirements.

General issues that affect software



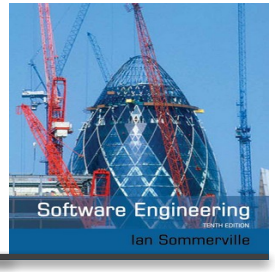
- Heterogeneity
  - Increasingly, **systems are required to operate as distributed systems** across networks that include different types of computer and mobile devices.
- Business and social change
  - Business and society are changing incredibly quickly as emerging economies develop and new technologies become available. **They need to be able to change their existing software and to rapidly develop new software**.

General issues that affect software



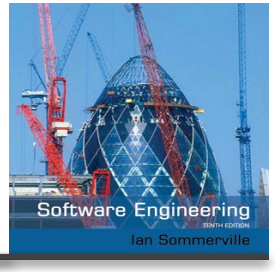
- Security and trust
  - As software is intertwined with all aspects of our lives, it is essential that we can trust that software.
- Scale
  - Software has to be developed across a very **wide range of scales**, from very small **embedded systems in portable** or wearable devices through to Internet-scale, **cloud-based systems that serve a global community.**

Software engineering diversity



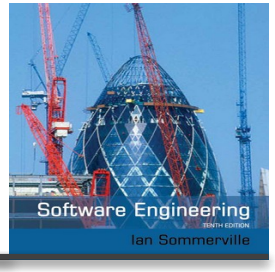
- There are **many different types of software system** and there is **no universal set of software techniques that is applicable to all of these.**
- The software engineering methods and tools used depend on the type of application being developed, the requirements of the customer and the background of the development team.

Application types



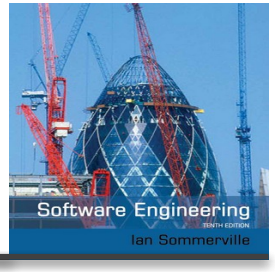
- Stand-alone applications
- Interactive transaction-based applications
- Embedded control systems
- Batch processing systems
- Entertainment systems
- Systems for modelling & simulation
- Data collection systems

Application types



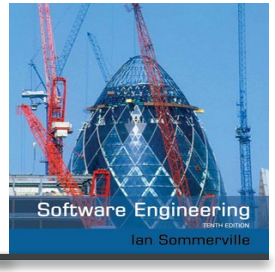
- Stand-alone applications
  - These are application systems that **run on a local computer**, such as a PC. They include all necessary functionality and do not need to be connected to a network.
- Interactive transaction-based applications
  - Applications that execute on a remote computer and are accessed by users from their own PCs or terminals. These **include web applications such as e-commerce applications**.
- Embedded control systems
  - These are **software control systems that control and manage hardware devices**. Numerically, there are probably more embedded systems than any other type of system.

Application types



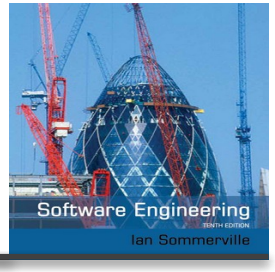
- Batch processing systems
  - These are business systems that are **designed to process data in large batches**. They process large numbers of individual inputs to create corresponding outputs.
- Entertainment systems
  - These are systems that are primarily for personal use and which are **intended to entertain the user.**
- Systems for modelling and simulation
  - These are systems that are developed by scientists and engineers to **model physical processes or situations, which include many, separate, interacting objects.**

Application types



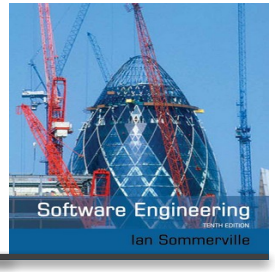
- Data collection systems
  - These are systems that **collect data from their environment using a set of sensor**s and send that data to other systems for processing.

Software engineering fundamentals



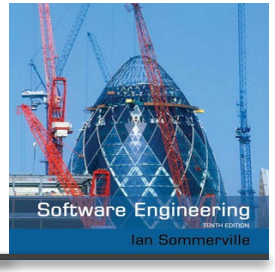
- Some fundamental principles apply to all types of software system, irrespective of the development techniques used:
  - **Systems** should be **developed using software engineering methodologies.**
  - The system should **perform efficiently.**
  - **Understand & implement requirements clearly.**
  - **Reuse software** rather than creating from scratch.

Internet software engineering



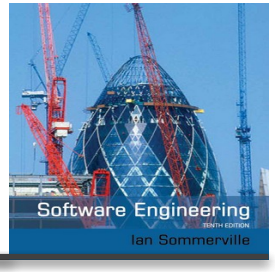
- **The Web is now a platform for running applications** and organizations are increasingly developing web-based systems rather than local systems.
- Web services allow application functionality to be accessed over the web.
- **Cloud computing** is an approach to the provision of computer services where applications run remotely on the ‘cloud’.

Web-based software engineering



- Web-based systems are complex distributed systems.
- All fundamental principles discussed earlier are followed by web based systems as well.

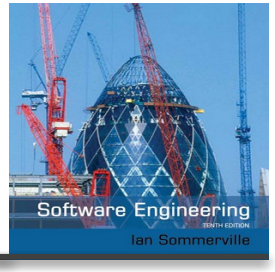
Web software engineering



- Software reuse
  - Mostly preferred approach for constructing web-based systems.
  - **When building these systems,**
    - **Think of assembling from pre-existing software components and systems.**

- Incremental and agile development
  - Web-based systems should be **developed and delivered incrementally.**
  - Because its impractical to specify all the requirements for such systems in advance.

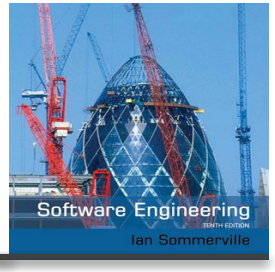
Web software engineering



- Service-oriented systems
  - Build software components using web services rather than building it from scratch.
  - For example,
    - Design a real-world parking application with all components from the scratch like one for login or signup another containing the map, and many more.
    - Or I can use google maps for parking availability, pay using payPal or login using Facebook etc. These all are independent web services.
- Rich interfaces
  - Interface development technologies such as AJAX and HTML5 have emerged that support the creation of rich interfaces within a web browser.

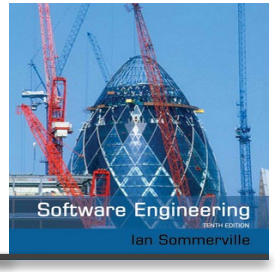
Software engineering ethics

Software engineering ethics



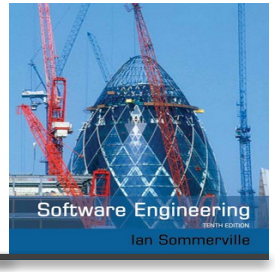
- Software engineering involves wider responsibilities than simply the application of technical skills.
- **Software engineers must behave in an honest and ethically responsible way** if they are to be respected as professionals.
- Ethical behaviour is more than simply upholding the law but involves following a set of principles that are morally correct.

Issues of professional responsibility



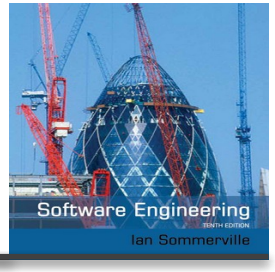
- Confidentiality
  - Engineers should normally **respect the confidentiality of their employers or clients** irrespective of whether or not a formal confidentiality agreement has been signed.
- Competence
  - Honestly quote your level of competence.
  - & Never accept the work that is out of your competence

Issues of professional responsibility

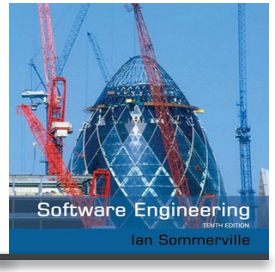


- Intellectual property rights
  - **Engineers should be aware of local laws governing the use of intellectual property such as patents, copyright**, etc.
  - They should be careful to ensure that the intellectual property of employers and clients is protected.
- Computer misuse
  - Software engineers should **not use their technical skills to misuse other people's computers.** Computer misuse ranges from relatively trivial (game playing on an employer's machine, say) to extremely serious (dissemination of viruses).

Ethical dilemmas

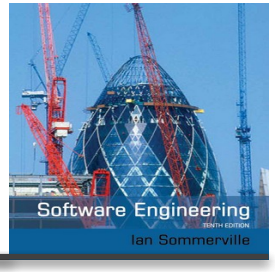


- Disagreement in principle with the policies of senior management.
- Your employer acts in an unethical way and releases a safety-critical system without finishing the testing of the system.
- Participation in the development of military weapons systems or nuclear systems. (ethically wrong to develop weapons but strategically correct)



Case studies

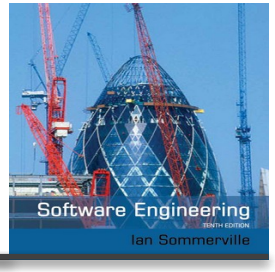
Case studies



- A personal insulin pump
  - **An embedded system** in an insulin pump used by diabetics to maintain blood glucose control.
- A mental health case patient management system
  - Mentcare. **A system used to maintain records** of people receiving care for mental health problems.
- A wilderness weather station
  - **A data collection system** that collects data about weather conditions in remote areas.
- iLearn: a digital learning environment
  - A system to support learning in schools.
  - A LMS

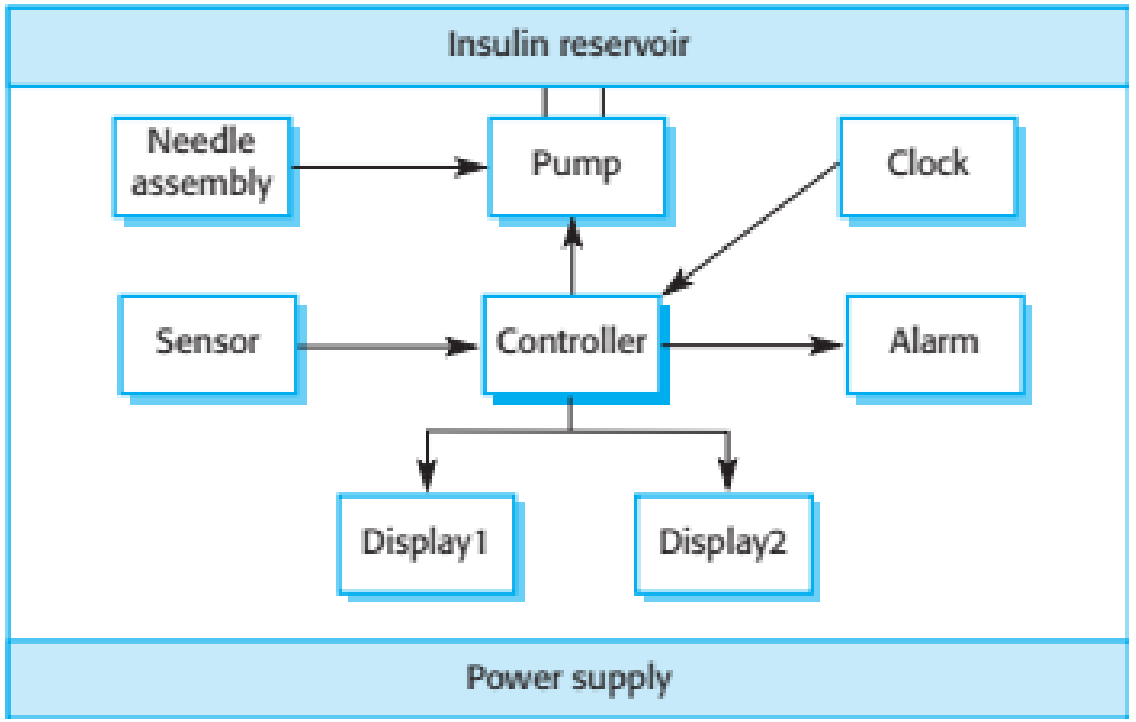
Personal Insulin Pump

Insulin pump control system

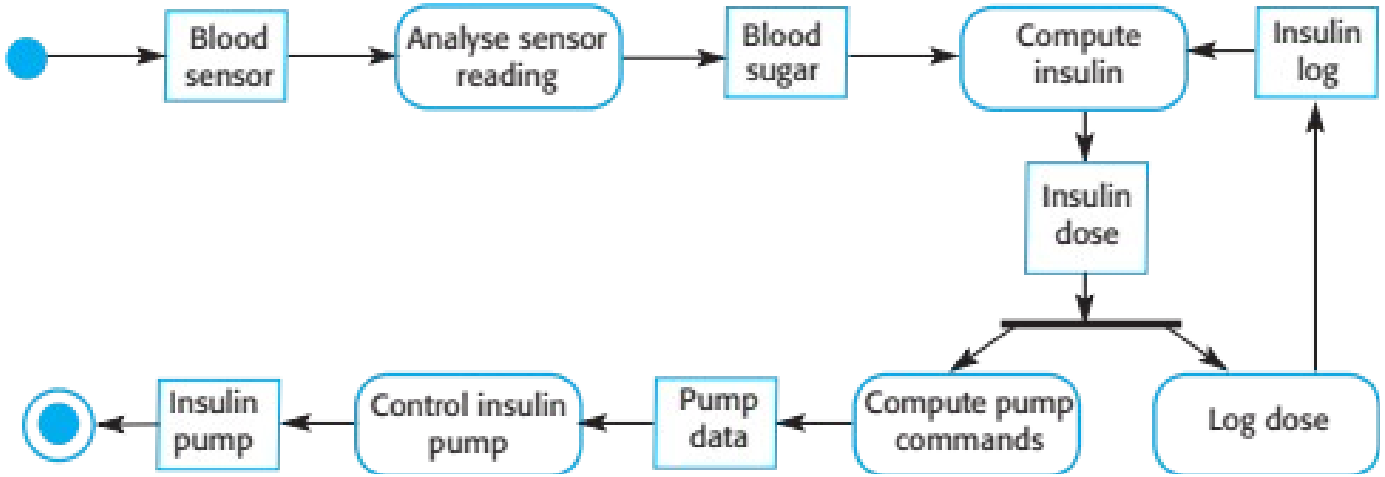


- **Working Steps:**
  - Collects data from a blood sugar sensor
  - calculates the amount of insulin required to be injected.
    - Calculation on the basis of rate of change of blood sugar levels.
  - Sends signals to a micro-pump to deliver the correct dose of insulin.
- **Safety-critical system**
  - <u>as low blood sugars</u>
    - can lead to brain malfunctioning, coma and death;
  - <u>high-blood sugar levels</u>
    - have long-term consequences such as eye and kidney damage.

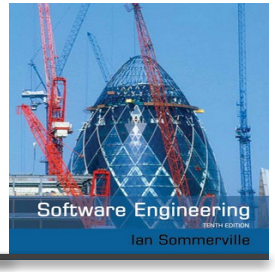
Insulin pump hardware architecture



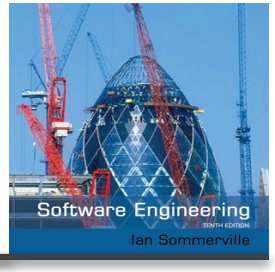
Activity model of the insulin pump



Essential Non-Functional requirements

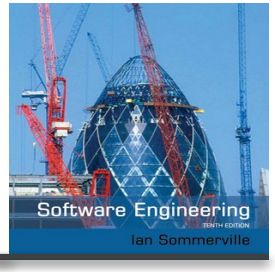


- **Availability:**
  - The system shall be available to deliver insulin when required.
- **Reliability:**
  - The system shall perform reliably and deliver the correct amount of insulin to counteract the current level of blood sugar.
- **Correctness:**
  - The system must therefore be designed and implemented to ensure that the system always meets these requirements.



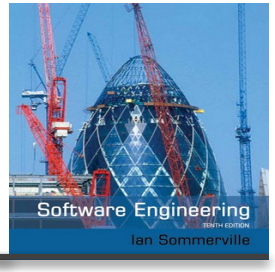
Mentcare: A patient information system for mental health care

Mentcare: A patient information system for mental health care



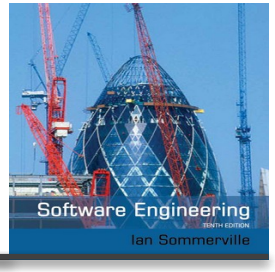
- A system that is designed for storing the mental health of patients is a medical **information system**
  - that maintains information about patients
  - and the treatments that they have received.
- **Reason for development:**
  - Most mental health patients need to attend specialist clinics regularly where they can meet a doctor who has detailed knowledge of their problems.
  - **Mentcare** is an information system that is **intended for use in clinics**.

Mentcare



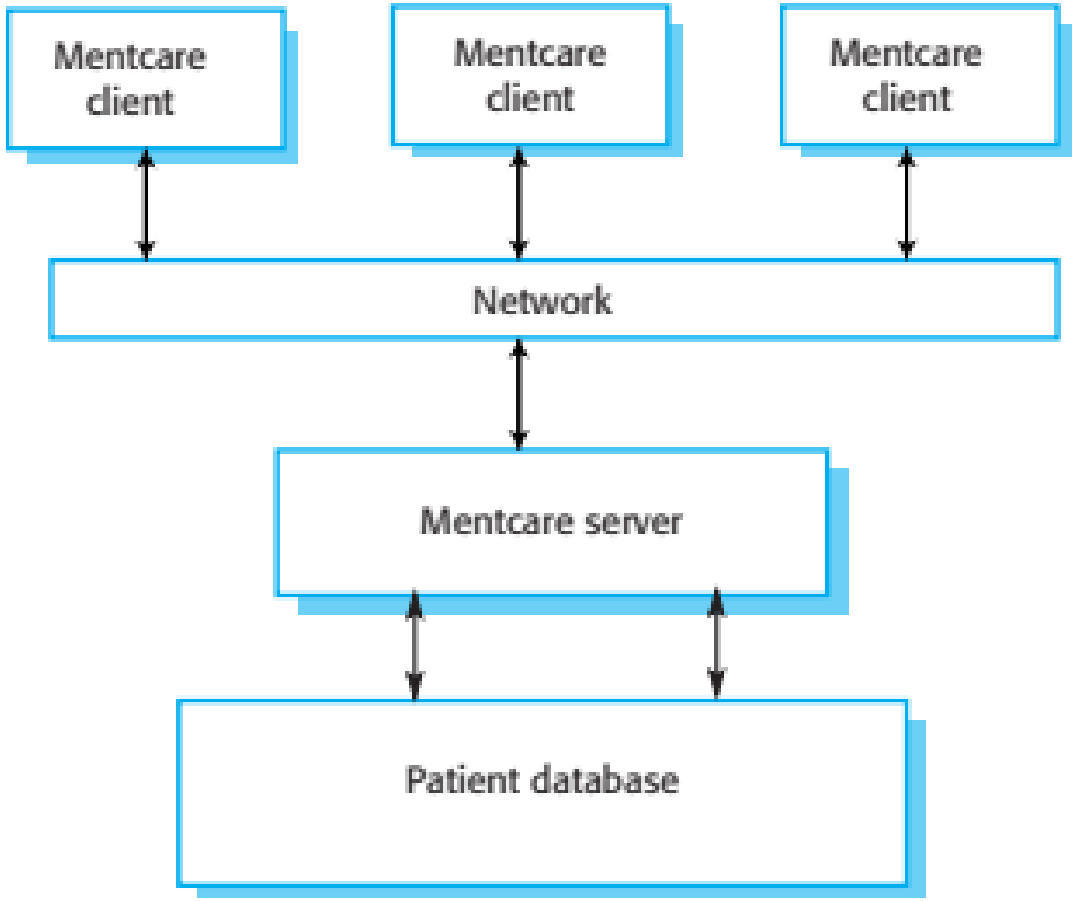
- **KEY components:**
  - **A centralized database** of patient information but has also been designed to run on a PC.
  - The patient information in the database can be downloaded so that it can be accessed offline.

Mentcare goals

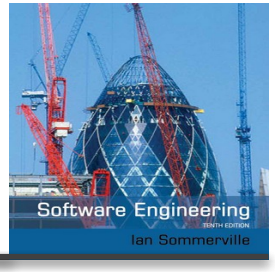


- allows health service managers to assess performance against local and government targets.
- provide timely information to medical staff for ease in treating of patients.

The organization of the Mentcare system

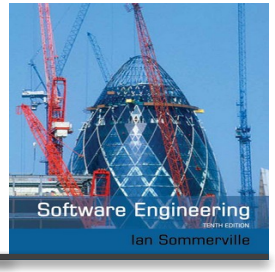


Key features of the Mentcare system



- **Individual care management**
  - Clinicians can create records for patients,
  - edit the information in the system,
  - view patient history,
  - show data summaries for doctors.
- **Patient monitoring**
  - The system monitors the records of patients that are involved in treatment and issues warnings, if possible, problems are detected.
- **Administrative reporting**
  - The system generates monthly management reports showing the number of patients treated at each clinic, the number of patients who have entered and left the care system, number of patients sectioned, the drugs prescribed and their costs, etc.

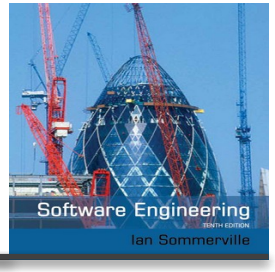
Mentcare system concerns



- **Privacy**
  - It is essential that patient information is confidential and is never disclosed to anyone apart from authorised medical staff and the patient themselves.
- **Safety**
  - Some **mental illnesses cause patients to become suicidal or a danger to other people. Wherever possible, the system should warn medical staff about potentially suicidal or dangerous patients.**
  - The system must be available when needed otherwise safety may be compromised and it may be impossible to prescribe the correct medication to patients (depending upon history log).
- Req. conflict
  - As privacy can be maintained by keeping 1 copy of record but availability can be achieved by data redundancy.
  - A trade-off between two NFRs.

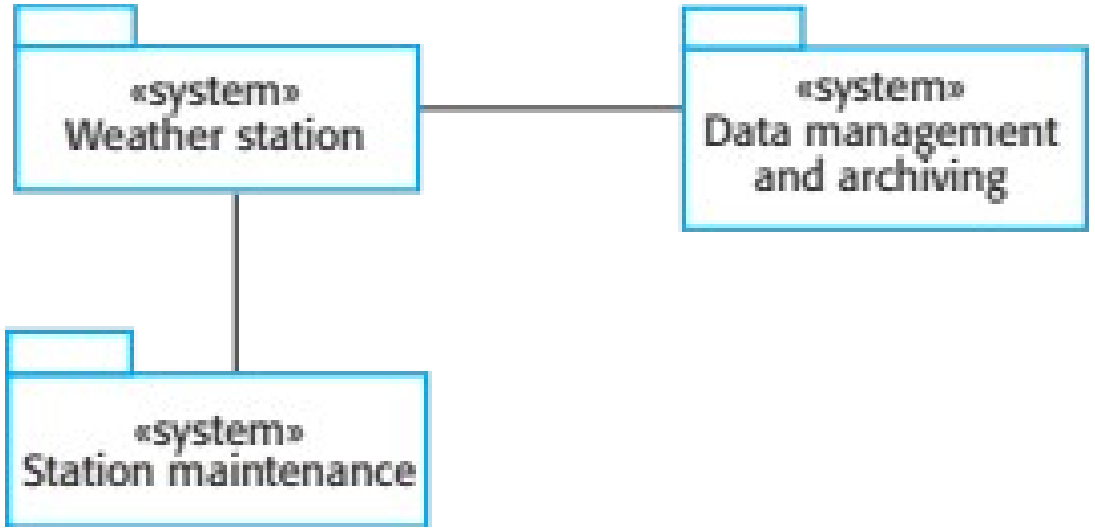
Wilderness weather Station

Wilderness weather station

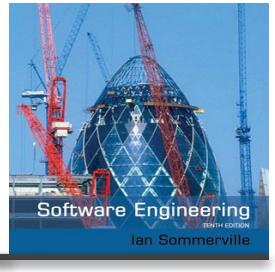


- A data collection system
- The government of a country with large areas of wilderness decides to deploy several hundred weather stations in remote areas.
- Weather stations collect data from a set of instruments that measure temperature and pressure, sunshine, rainfall, wind speed and wind direction.
  - The weather station includes a **number of instruments** that **measure** weather parameters such as the **wind speed and direction, the ground and air temperatures, the barometric pressure and the rainfall over a 24-hour period**. Each of these instruments is controlled by a software system that takes parameter readings periodically and manages the data collected from the instruments.

The weather station’s environment

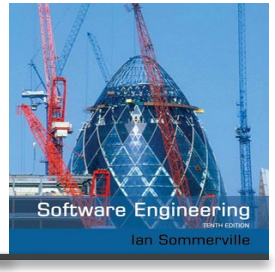


Weather information system



- The weather station system
  - This is responsible for **collecting weather data**, carrying out some **initial data processing,** and **transmitting it to the data management system.**
- The data management and archiving system
  - This system **collects the data from all of the wilderness weather stations,** carries out data processing and analysis and archives the data.
- The station maintenance system
  - This system can **communicate by satellite with all wilderness weather stations to monitor the health of these systems** and provide reports of problems.

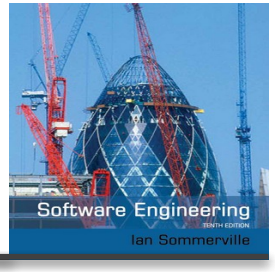
Additional software functionality



- **Monitor the instruments,** power and communication hardware and report faults to the management system.
- **Manage the system power**, ensuring that batteries are charged whenever the environmental conditions permit but also that generators are shut down in potentially damaging weather conditions, such as high wind.
- **Support dynamic reconfiguration** where parts of the software are replaced with new versions and where backup instruments are switched into the system in the event of system failure.

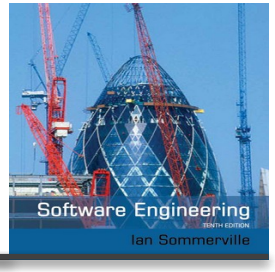
iLearn: a Digital learning Environment

iLearn: A digital learning environment



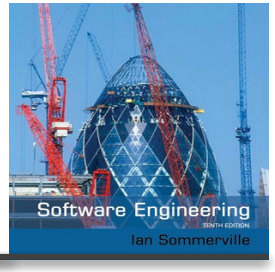
- A digital learning environment is a framework in which a set of general-purpose and specially designed tools for learning may be embedded plus a set of applications that are geared to the needs of the learners using the system.
- The tools included in each version of the environment are chosen by teachers and learners to suit their specific needs.
  - These can be general applications such as spreadsheets, learning management applications such as a Virtual Learning Environment (VLE) to manage homework submission and assessment, games and simulations.

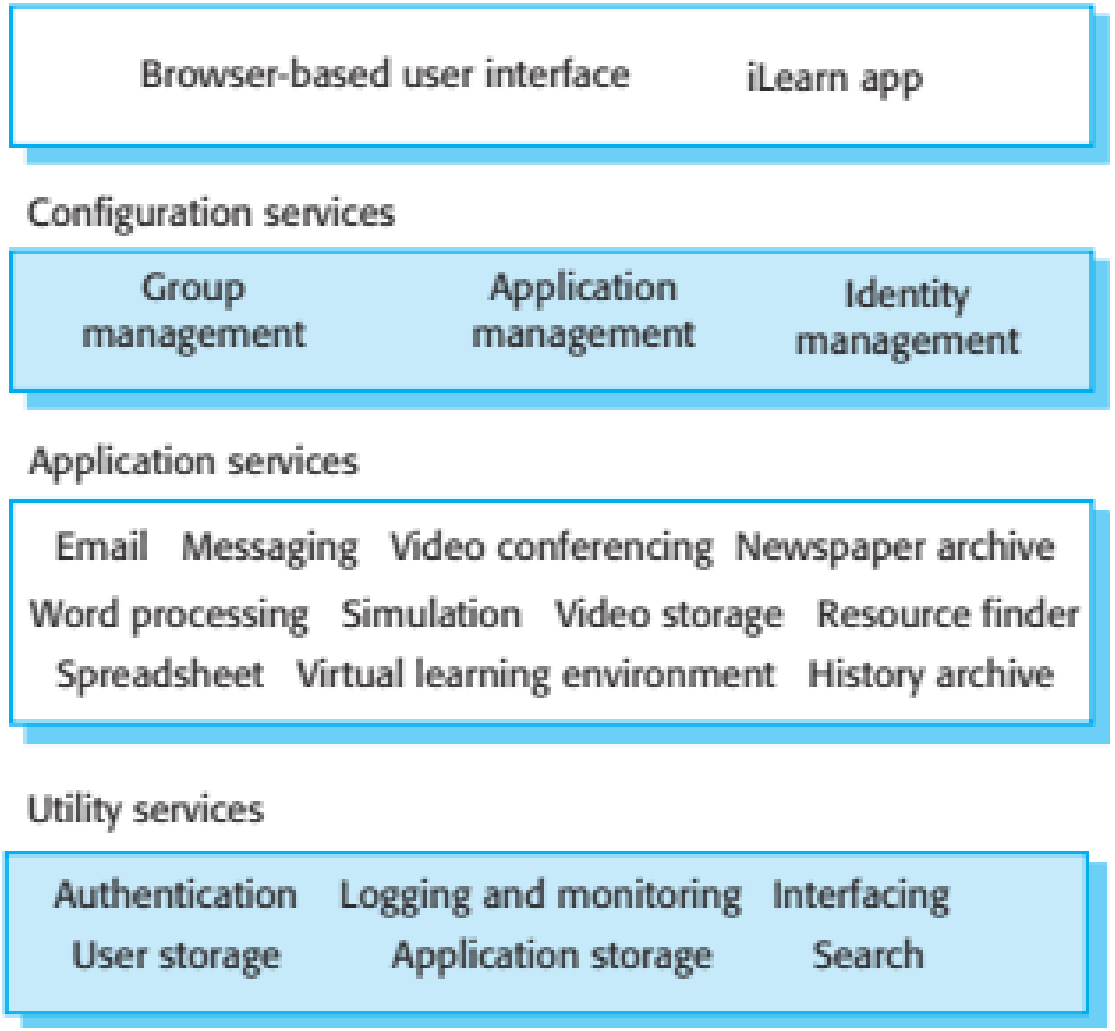
Service-oriented systems



- The system is a **service-oriented system** with all system components considered to be a replaceable service.
- This **allows the system to be updated incrementally** as new services become available.
- It also makes it possible to rapidly configure the system to create versions of the environment for different groups such as very young children who cannot read, senior students, etc.

iLearn services



- **Utility services**
  - that provide basic application-independent functionality, and which may be used by other services in the system.
- **Application services**
  - that provide specific applications such as email, conferencing, photo sharing etc. and access to specific educational content such as scientific films or historical resources.
- **Configuration services**
  - that are used to adapt the environment with a specific set of application services and do define how services are shared between students, teachers and their parents.

iLearn architecture